

MODULE 1- SUSTAINABILITY

Syllabus Covered:

- ❖ Sustainability- Introduction
- ❖ Sustainability- Definition
- ❖ Sustainability- Need & Concept
- ❖ Social Sustainability Concept
- ❖ Environmental Sustainability Concept
- ❖ Economic Sustainability Concept
- ❖ Sustainable Development
- ❖ Nexus Between Technology And Sustainable Development
- ❖ Challenges to Sustainable Development
- ❖ Multilateral Environmental Agreements And Protocols
- ❖ Clean Development Mechanism (CDM)
- ❖ Environmental Legislations In India – Air Act, Water Act

1. SUSTAINABILITY- INTRODUCTION

The concept of sustainability become so important nowadays, because of the irreparable damage caused to the environment by industrial civilization & consumerism, which originated about 3 centuries ago. They were based on the following wrong assumptions:

- Earth belongs to the humans only
- Ignore the fact that humans are part of the Earth's biosphere
- Earth's stock of resources are infinite
- Environment can bear any amount of damage that is caused by human activity.

The advantages & disadvantages of industrial civilization & consumerism are listed below:-

Advantages

1. Flourished the economy
2. Improved the living quality of the society

Disadvantages

1. Caused irreparable damage to the environment
2. Pose a threat to the life support systems of earth.
3. E.g.1.The hole in the ozone layer surrounding our planet as a shield against the dangerous ultraviolet radiation from the sun
4. E.g.2. Increased carbon dioxide content and greenhouse gases in the atmosphere that cause global warming

Several warnings concerning the instability of Earth's life support systems have been raised in the recent times. In 1992, some of the world's senior scientists from 70 countries, signed and sent an urgent warning "The environment is suffering critical stress..." to the government leaders of all nations as part of the *United Nations Conference on Environment and Development* (the "Earth Summit") held in Rio de Janeiro, Brazil. This marked the beginning of the concept sustainability.

Sustainability is based on a simple principle: Everything that we need for our survival and well-being depends, either directly or indirectly, on our natural environment. Sustainability creates and maintains the conditions under which humans and nature can exist in productive harmony, that permit fulfilling the requirements of present and future generations. Sustainability is important to making sure that we have and will continue to have, the water, materials, and resources to protect human health and our environment.

2. SUSTAINABILITY- DEFINITION

Sustainability is the ability to achieve continuing economic prosperity while protecting the natural systems of the planet and providing a high quality of life for its people.

3. SUSTAINABILITY- NEED & CONCEPT

Sustainability has three components, which are inter-related, as shown in Fig.1:

1. Environment
2. Society
3. Economy

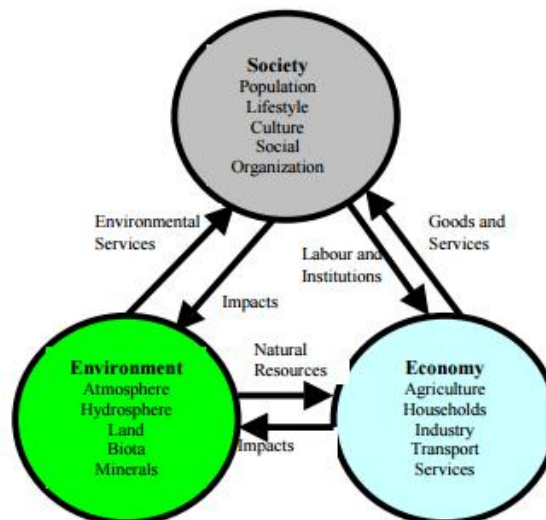


Fig.1. Environment-Economy-Society Inter-relation

Environment gives resources, raw materials to the Economy for production activities. Economy creates products and sells it to society for use. Production by Economy and Consumption by Society lead to the following environmental impacts.

1. Exhaustion of Resources – Water, Petroleum, Forests
2. Loss of Biodiversity - Extinction of Animal/Plant Species due to Water, Soil, Air Pollution
3. Deforestation - conversion of forestland to farms, urban use etc.
4. Ozone Depletion - reduction of the amount of ozone in the stratosphere due to the emission of chlorofluorocarbons (CFCs). CFC/s emitted from the industries, rises to the Stratosphere. Sunlight breaks CFCs to release Chlorine. Chlorine reacts with Ozone and destroys it.
5. Acid Deposition – results in acid rain, acid fog and acid mist.
6. Desertification - type of land degradation in which a land region becomes dry, typically losing its water bodies, vegetation and wildlife.
7. Eutrophication - form of water pollution occurs when excessive fertilizers run into lakes and rivers. This encourages the overgrowth of algae and other aquatic plants.
8. Global Warming - gradual increase in the average temperature of the Earth's atmosphere and its oceans, caused by increasing concentrations of greenhouse gases – Carbon oxides, Nitrous oxides, sulphur oxides, Fluorocarbons

The environmental impact, caused by economy on production and society on consumption, leads to the following damages to human life.

1. Fresh water scarcity
2. Climate change
3. Exposure to **toxics** in food, air, water and soil
4. Emerging diseases
5. Food insecurity resulting in poverty
6. **Energy** scarcity due to depletion of **non-renewable resources**
7. Ecosystem damage and habitat loss due to **pollutant** discharges
8. Sea level rise

The need of sustainability is to reduce these damages and create a livable planet earth for the future generations. For this, United Nations presented the following key sustainability concepts:- **Inter-generational equity** – Expects the present generation to hand over a safe, healthy and resourceful environment to the future generation. **Intra-generational equity** – Emphasize the technological development should support economic growth of the poorer section, so as to reduce the gap between nations. Sustainability means balancing environment, society and economy, as shown in Fig 2.

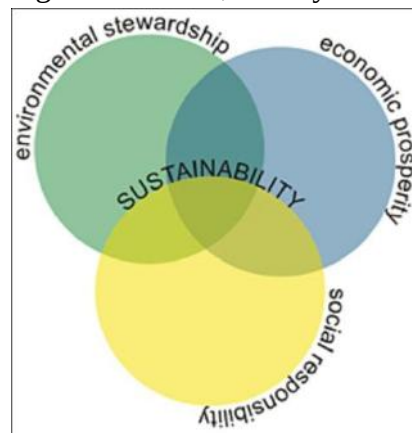


Fig 2: Sustainability

4. SOCIAL- ENVIRONMENTAL AND ECONOMIC SUSTAINABILITY CONCEPTS

The concept of sustainability is based on the basis that people and their communities are made up of social, economic, and environmental systems that are in constant interaction and that must be kept in harmony.



SOCIAL SUSTAINABILITY

There are six principles of sustainability that can help a community ensure that its social, economic, and environmental systems are well integrated and will endure. A community or society that wants to pursue sustainability will try to:

1. Maintain residents' quality of life.

Quality of life has many components: income, education, health care, housing, employment, legal rights. Each locality must define and plan for the quality of life it wants and believes it can achieve, for now and for future generations.

2. Enhance local economic vitality.

A viable local economy is essential to sustainability. This includes job opportunities, sufficient tax base and revenue to support government and the provision of infrastructure and services, and a suitable business climate.

3. Promote social and intergenerational equity.

A sustainable community's resources and opportunities are available to everyone, regardless of ethnicity, age, gender, cultural background, religion, or other characteristics. Further, a sustainable community does not deplete its resources and destroy natural systems.

4. Maintain the quality of the environment.

A sustainable community tries to find ways to co-exist with natural environment and ecosystem. It avoids unnecessary degradation of the air, oceans, fresh water, and other natural systems.

5. Incorporate disaster resilience and mitigation into its decisions and actions.

A community is resilient in the face of inevitable natural disasters like tornadoes, hurricanes, earthquakes, floods, and drought if it takes steps to ensure that such events cause as little damage as possible.

6. Use a consensus-building, participatory process when making decisions.

Participatory processes are vital to community sustainability... It encourages the identification of concerns and issues, promotes the wide generation of ideas for dealing with those concerns, and helps those involved find a way to reach agreement about solutions.

ENVIRONMENTAL SUSTAINABILITY

Environmental sustainability requires:

1. Maintenance of biodiversity (genes, species and ecosystems)
2. Protection of natural capital (air, water, soils etc)
3. Maintenance of the energy and material cycles of the planet
4. Health and resilience of all life support systems.

This can be achieved by:

1. Reduce dependence upon finite, virgin resources like Fossil fuels, minerals and metals
2. Nature must not be subjected to increased concentrations of substances produced by society. This requires that consideration be given to the biodegradability of substances and the length of time it takes the earth to reabsorb them.
3. The physical basis for the productivity and biodiversity of nature must be not systematically degraded. This requires that we protect diverse and special habitats.
4. There must be efficient use and fair distribution of resources to enable humans to meet their needs. This requires a reduction in consumerism, especially among wealthy nations.

ECONOMIC SUSTAINABILITY

The economic sustainability ensures that the industry or business is making profit without creating much damage to environment/ecology. Economic growth is expressed in terms of Gross Domestic Product (GDP). This is the total amount of production produced within a nation, within one year. Economic growth has to be sustainable, if it improves quality of human life. Thus population factor must be included to ensure fair resource consumption.

	Developed Country	Developing Country
Resource Consumption	Not Sustainable	Sustainable
Population	Sustainable	Not Sustainable

ECONOMIC-SOCIAL AND ENVIRONMENTAL MATRIX

Quality of Life Concerns	Economic Issue		Social Issue		Environmental Issue	
	Unsustainable	Sustainable	Unsustainable	Sustainable	Unsustainable	Sustainable
Water	High cost of drinking water	Drinking water availability at low cost	Access to drinking water denied to weaker section	Adequate water supply to all sections	High-level of pollution in lakes and rivers	Conservation of existing fresh water bodies
Food	High cost of food and use of fertilizers and pesticides in farming	Good food available at low cost	Access to good food denied to weaker section	Adequate access to good food to all sections	Overuse of fertilizers and pesticides pollute the environment. Deforestation - conversion of forestland to farms	Food is of nutritious quality and related diseases are lowered.
Energy	High cost and intermittent power supply	Electricity available at low cost	Overuse of energy by the rich society and inadequate energy distribution	Adequate energy available to all sections	Use of fossil fuels and pollution	Use of renewable resources.(solar, wind, biomass)

5. SUSTAINABLE DEVELOPMENT

The concept of sustainable development has received much recognition after the Stockholm declaration in the year 1972.

Sustainable development is the development which meets the needs of the present without compromising the ability of future generations to meet their own needs. (Definition proposed by the Brundtland Commission in 1987 in their report “Our Common Future”).

The three pillars of sustainable development are environment, society and economy as shown in Fig.3.

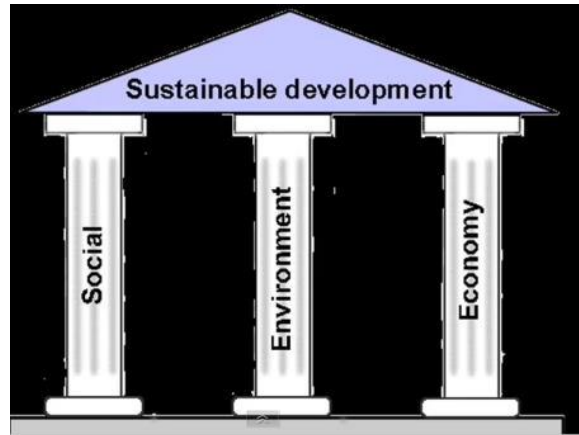


Fig. 3. Pillars of Sustainable Development

Sustainable development should have the following features:-

1. Satisfying human needs
2. Favouring a good quality of life through decent standards of living
3. Sharing resources between rich and poor
4. Acting with concern for future generations
5. Looking at the 'cradle-to-grave' impact when consuming
6. Minimizing resource use, waste and pollution

WHAT IS TO BE SUSTAINED?	WHAT IS TO BE DEVELOPED?
Nature – Earth, Biodiversity, Ecosystems	People – Child survival, Life expectancy, Education, Equity, Equal opportunity
Life support – Resources, Environment	Economy – Wealth, Production, Consumption
Community – Cultures, Places	Society – Institutions, Social capital, States, Regions

MEASURES OF SUSTAINABLE DEVELOPMENT

The following are the measures of sustainability development:-

(i) Technology:

Using appropriate technology is one which is locally adaptable, eco-friendly, cost effective, resource efficient and culturally suitable. Nature is often taken as a model, using the natural conditions of that region as its components. This concept is known as “design with nature”.

(ii) Reduce, Reuse, and Recycle Approach:

The 3-R approach advocating minimization of resource use, using them again, and recycling the materials. It reduces pressure on our resources as well as reduces waste generation and pollution.



(iii) Promoting Environmental Education and Awareness:

Making environmental education the centre of all learning process will greatly help in changing the thinking pattern and attitude of people towards our earth and the environment.

(iv) Resource Utilization as Per Carrying Capacity:

Any system can sustain a limited number of organisms on a long-term basis which is known as its carrying capacity. If the carrying capacity of a system is crossed (say, by over exploitation of a resource), environmental degradation starts.

(v) Improving Quality of Life Including Social, Cultural and Economic Dimensions:

Development should not focus just on one-section of already affluent people. Rather it should include sharing of benefits between the rich and the poor. The tribal, ethnic people and their cultural heritage should also be conserved.

6. NEXUS BETWEEN TECHNOLOGY AND SUSTAINABLE DEVELOPMENT

Technology is the offspring of science. Technological innovation can be seen as a 'double edged sword', with respect to sustainable development.

1. Technology improves quality of life, eliminate diseases and increase life expectancy
2. On the other hand, technology creates irreparable environmental damage due to resource extraction and pollution of air, water, soil.

As technology advances, the environmental degradation accelerates exponentially. Also the benefits of technological innovations are mostly enjoyed by the developed countries. The technology remains as a dream for underdeveloped countries which still face poverty, inadequate sanitation facilities etc. Hence it is essential to integrate technology, society into sustainability.

Technology can support sustainability by

1. Conserving natural capital (renewable and nonrenewable resources)
2. Reducing waste and pollution
3. Raising efficiency standards
4. Finding substitutes for toxic/hazardous materials

Pollution prevention and cleaner production technologies are more cost effective than end of pipe waste treatment technology. Some of the technological applications towards sustainable development in various sectors are given below.

1. SUSTAINABLE AGRICULTURAL TECHNOLOGY

Sustainable agriculture integrates three main goals-environmental health, economic profitability, and social and economic equity. Some of the common ways towards sustainable agriculture are:

- a) Integrated Pest Management (IPM)
- b) Rotational Grazing
- c) Soil conservation
- d) Water quality/wetlands
- e) Cover crops
- f) Crop/ landscape diversity
- g) Nutrient management
- h) Agro-forestry

2. SUSTAINABLE ENERGY

Sustainable energy is the energy that, in its production or consumption, has minimal negative impacts on human health and the healthy functioning of vital ecological systems, including the global environment. This can be achieved by using the following:

- a. Renewable energy sources
 - i. Solar
 - ii. Biomass (It is a renewable energy resource derived from the carbonaceous waste of various human and natural activities. It is derived from numerous sources, including the by-products from the timber industry, agricultural crops, raw material from the forest, major parts of household waste and wood.)
 - iii. Wind
 - iv. Tide
 - v. Geothermal Heat
- b. Energy efficient systems - upgrading the efficiency of the existing equipment, reduction of energy loss, saving of fuel, and optimization of its operating conditions and service life provide an ecologically safe strategy.

7. CHALLENGES TO SUSTAINABLE DEVELOPMENT

The path of sustainable development can only lead us to the future. Some of the major challenges exist in the modern world that may hinder the effort towards sustainable development are given below:-

- 1. Misconception about sustainable development
 - The popular perception that development and environment protection should not go together, has to be changed.
- 2. Population explosion in developing economies (require more food, energy, goods)

3. Over exploitation of natural resources in developed countries (resulting in imbalance between developed & underdeveloped countries)
4. Poverty (To achieve sustainable development, eradication of poverty is vital.
5. Over dependence on fossil fuels leading to global warming & climate change
6. Loss in Biodiversity
7. Freshwater scarcity
8. High cost of appropriate technology to tap renewable energy sources
9. Poor solid waste management system (focusing on mere waste disposal)
10. Public awareness
11. Absence of strict environment laws and legislations to control pollution
12. Absence of adequate political and industrial will for moving towards a sustainable future
13. Lack of co-ordination between the three pillars of sustainable development
 - Economy, Society, Environment – in finding long term solutions
14. Corruption and misuse of sustainable development assistance funds

8. MULTILATERAL ENVIRONMENTAL AGREEMENTS AND PROTOCOLS

A multilateral environmental agreement (MEA) is a legally binding agreement between two or more countries containing commitments to meet specific environment-related objectives. MEA can be expressed as a Treaty, Convention, Agreement, Pact, Accord or Protocol. MEA is an international legal instrument that

- has environmental protection/sustainable development as their goal
- concluded between a large number of states or international organizations as parties
- concluded in written form
- governed by international law

The 1972 Stockholm Conference on the Human Development was the first attempt to address inter-relationships of environment and development at global level. It adopted first global action plan for environment and established the United National Environmental Programme (UNEP). MEAs have flourished in last three decades, especially after the 1972 International Stockholm Conference and the 1992 United Nations Conference on Environment and Development (UNCED), due to the following reasons:

- Response to the gravity of environmental problems
- Growing understanding of regional and global nature of environmental problems

MEA PROCESS

- MEAs are developed through a thorough negotiation process among concerned countries.
- The result is a mutually agreed upon treaty adopted via a high-level related conference, and signed by the signatories (states);
- “Party”/ country ratifies (accepts, approves or accedes) the convention
- After ratification of a fixed/ defined number of signatories, the convention enters into force
- Once a Party has finalized the ratification process; it is subject to all its provisions.

MEAs

- a. MEAs related to protection and conservation of biodiversity - CBD / Cartagena Protocol, CITES, RAMSAR
- b. MEAs focusing on protecting land from negative “altering”/land-use - UN Convention to Combat Desertification (UNCCD)
- c. MEAs aimed at protection of atmosphere from pollutants - Vienna Convention / Montreal Protocol on Substances that Deplete Ozone Layer, UN Framework Convention on Climate Change / The Kyoto Protocol
- d. MEAs relevant to marine environment - UNCLOS/Fish Stocks Agreement, Global Program of Action for the Protection of the Marine Environment from Land-Based Activities (GPA)
- e. MEAs deal with hazardous impacts of wastes and chemical pollution - Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal, Rotterdam Convention on the Prior Informed Consent for Certain Hazardous Chemicals and Pesticides in International Trade (PIC), Stockholm Convention on Persistent Organic Pollutants (POPs)

The Kyoto Protocol

The Kyoto Protocol, a legally binding global agreement by 192 countries, was adopted in Kyoto, Japan, on 11 December 1997 and entered into force on 16 February 2005. The Kyoto Protocol is an international agreement linked to the United Nations Framework Convention on Climate Change, which commits its Parties by setting internationally binding emission reduction targets. The Protocol places a heavier burden on developed nations – Australia, USA, European Countries, under the principle of “common but differentiated responsibilities”, recognizing that developed countries are principally responsible for the current high levels of GHG emissions in the atmosphere as a result of more than 150 years of industrial activity.

9. CLEAN DEVELOPMENT MECHANISM (CDM)

The Clean Development Mechanism is regarded as one of the most important internationally implemented market based mechanisms to reduce carbon emissions. Created under the Kyoto Protocol, the CDM was designed to help developed nations meet domestic greenhouse gas (GHG) reduction commitments by investing in low-cost emission reduction projects in developing countries.

The Clean Development Mechanism (CDM), established under the Kyoto Protocol, is the primary international offset program in existence today. It generates offset through investments in GHG reduction, and avoidance projects in developing countries. These offset credits, called Certified Emission Reduction credits (CERs), represent a reduction in one metric ton of carbon dioxide (CO₂) emitted to the atmosphere. Developed countries can use CERs to more cost-effectively achieve their Kyoto Protocol GHG emission reduction targets.

The stated purpose of the Clean Development Mechanism is to help developing countries achieve sustainable development, and assist industrialized countries in complying with their emission reduction commitments.

PURPOSE OF CLEAN DEVELOPMENT MECHANISM

Private companies fund projects in developing countries that reduce greenhouse gas emissions. They must also meet sustainable development criteria and the “additionality” requirement, which means the emission reductions made, must be “additional” to what would have been possible without CDM funding. Upon verification, the CDM awards these projects certified emission reductions (CERs), each equivalent to one ton of carbon dioxide. CERs are then sold to developed countries, which use them to meet a part of their reduction commitments under the Kyoto Protocol. CERs are also called “offset credits” because they “offset” the developed countries’ emissions with reductions in developing countries.

CDM allows countries to continue emitting green house gases, so long as they pay for reductions made elsewhere. The justification for this is based on the premise that it would be far more expensive to implement emission reduction in industrialized countries than in developing countries. It would help developing countries to gain sustainable development benefits from the entry of “clean” and more energy efficient technologies.

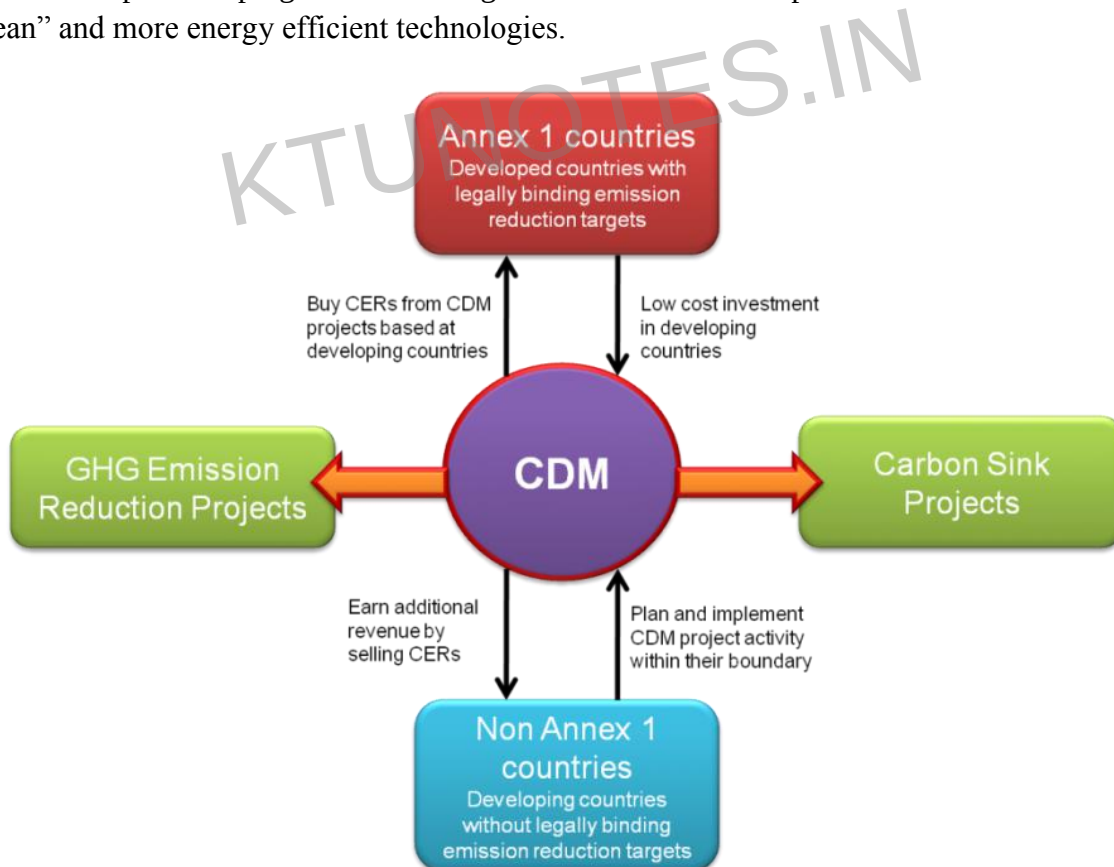


Fig: Clean Development Mechanism

10. ENVIRONMENTAL LEGISLATIONS IN INDIA

Laws that regulate the human interaction with environment are termed as Environmental laws. The environmental conservation is not a new concept to India, rather it is an inseparable part of Indian heritage and culture (Indians worship Panchabhootas – Earth, Water, Air, Fire, Space). Realizing the importance of environment, our country has made the following provisions in our Constitution.

1. Article 21 – Protection of life and personal liberty
2. Article 48 A – Protection and improvement of environment and safeguarding of forests and wildlife
3. Article 51 A – Fundamental duties of every citizen of India – To Protect and improve the natural environment including forests, rivers, lakes and wildlife

The following are the major environment legislations of India.

1. Water (Prevention & Control of Pollution) Act 1974
2. Air (Prevention & Control of Pollution) Act 1981
3. Environment Protection Act 1986 (May 1986)
4. Environment Protection Rules 1986 (Nov 1986)

10.1 WATER (PREVENTION AND CONTROL OF POLLUTION) ACT OF 1974

This is an act to provide measures for the prevention and control of water pollution and maintaining or restoring the wholesomeness of water. The act prohibits the disposal of any poisonous, noxious or polluting matter or any matter causing obstruction to proper flow of water in a stream.

Some of the salient features of Water act are listed below:-

- a. Establishment of pollution control board at Central (Central Pollution Control Board) and State level (State Pollution Control Board)
- b. Mandates prior approval of Pollution Control Board for operating, establishing and expanding industrial activities leading to discharge of industrial effluents
- c. Empowers Pollution Control Board to enter and inspect plant, factory etc
- d. Empowers Pollution Control Board to take samples of industrial effluents and do subsequent analysis
- e. Advise the Central/State Government on any matter concerning the prevention, control or abatement of water pollution.

10.2 AIR (PREVENTION AND CONTROL OF POLLUTION) ACT -1981

The Act was passed under the article 253 of the constitution of India which provide measures for the prevention, control and abatement of air pollution.

Air pollution is defined as the presence of any liquid or gaseous substances in the atmosphere in such concentration as may be or tend to be injurious to human being or other living creatures or plants or property or environment.

Some of the salient features of Air act are listed below:-

- a. Empowers Pollution Control Board to take appropriate measures for the prevention, control and abatement of air pollution.
- b. Mandates prior approval of Pollution Control Board for operating, establishing and expanding industrial activities that lead to discharge of air pollutants to the atmosphere

- c. Empowers Pollution Control Board to enter and inspect plant, factory etc
- d. Empowers Pollution Control Board to take samples of ambient air, emissions and do subsequent analysis
- e. Advise the Central/State Government on any matter concerning the prevention, control or abatement of air pollution.

IMPORTANT QUESTIONS

1. Define Sustainability.
2. Write a short note on the need of sustainability.
3. Explain sustainability with respect to social, economical, environmental concept.
4. Define Sustainable Development.
5. Explain the link between Sustainable Development and Technology.
6. What is MEA? Name some MEAs.
7. Define the process of MEAs.
8. What is Kyoto Protocol?
9. What are the challenges of Sustainable Development?
10. Explain the purpose of Clean Development Mechanism.
11. What are the salient features of Water Act?
12. What are the salient features of Air Act?

PROJECT (ASSIGNMENT)

- a) Identifying/assessment of sustainability in your neighbourhood in education, housing, water resources, energy resources, food supplies, land use, environmental protection etc.
- b) Identify the threats for sustainability in any selected area and explore solutions for the same.